EN5102 – Digital Systems Design

# Assignment 2- System Bus Architecture

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# Assignment -2: 8-bit Parallel Bus Design

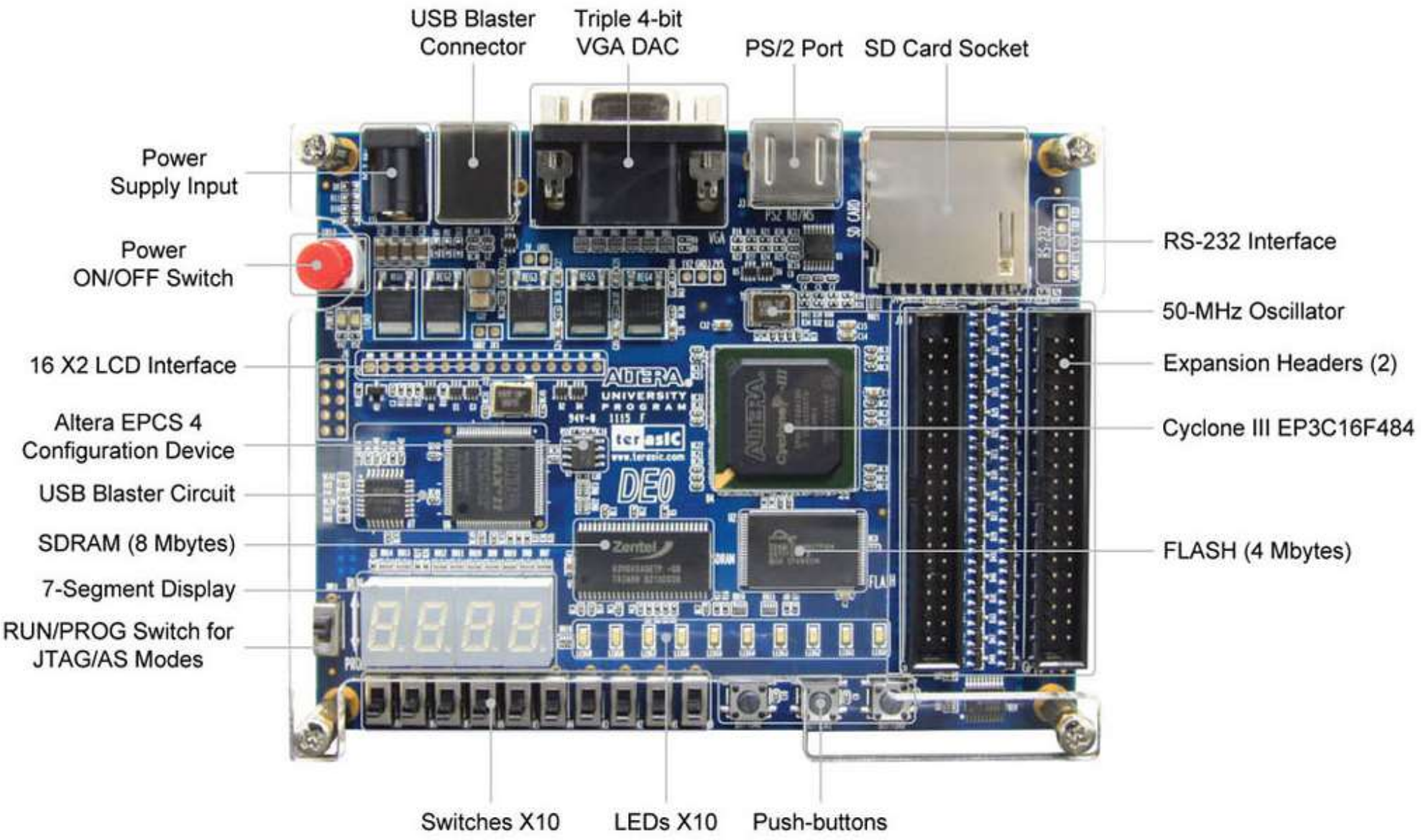
- Top level design: Coding and Integration
- 2 Masters & 3 slaves with 4K, 4K & 2K memory space
- Write your RTL with async reset & posedge clock
- **Implement your bus design in one DE0 board and use another DE0 board to implement your Master and slave models.**

| **Task** | **Notes:** |
|---|---|
| 1. *Arbiter Design* | Priority based arbitration. |
| 2. *Arbiter Verification* | Idle, Single master request, and two master requests |
| 3. *Address decoder* | Address decoder verification, 3 slaves, address mapping |
| 4. *Top level verification* | a) Reset test, b) Idle condition, c) one master request, d) two master requests |

**Note:** Your report is evaluated for 20 marks, and your demo carries 20 marks. Your report should have functional descriptions, IO definition, timing diagrams, RTL & Testbench codes.

**Deadline: 15 Days from Today**.

# Altera – DE0 Board



## Assignment 3 - Guidelines

1. Use one push-button for reset.
2. Use the 7-segment display to define your operation mode.
   1- idle, 2- one master request, 3- two master requests
3. Use 4 switches to set Address[3:0], and other MSB bits can be set to 0.
4. Use 4 switches to set Data[3:0], and other MSB bits can be set to 0.